

Applying (5.39) with $M = 24$, we see that the maximum error in the approximation is not greater than

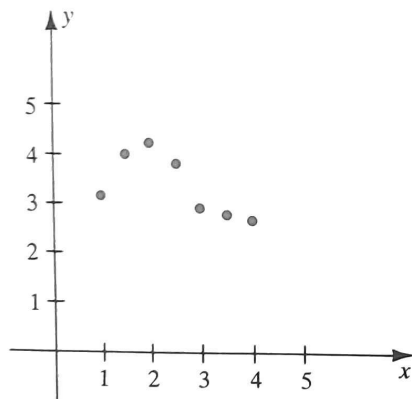
$$\frac{24(2-1)^5}{180(10)^4} = \frac{2}{150,000} < 0.00002.$$

Note that this estimated error is much less than that obtained using the trapezoidal rule in Example 1.

An important aspect of numerical integration is that it can be used to approximate the definite integral of a function that is described by means of a table or graph. To illustrate, suppose it is found experimentally that two physical variables x and y are related as shown in the following table

x	1.0	1.5	2.0	2.5	3.0	3.5	4.0
y	3.1	4.0	4.2	3.8	2.9	2.8	2.7

FIGURE 5.37



The points (x, y) are plotted in Figure 5.37. If we regard y as a function of x , say $y = f(x)$ with f continuous, then the definite integral $\int_1^4 f(x) dx$ may represent a physical quantity. In the present illustration, the integral may be approximated without knowing an explicit form for $f(x)$. In particular, if the trapezoidal rule (5.36) is used, with $n = 6$ and $(b - a) = (4 - 1)/12 = 0.25$, then

$$\int_1^4 f(x) dx \approx 0.25[3.1 + 2(4.0) + 2(4.2) + 2(3.8) + 2(2.9) + 2(2.8) + 2.7]$$

or

$$\int_1^4 f(x) dx \approx 10.3.$$

The number of subdivisions is even, so we could also approximate the integral by means of Simpson's rule.

EXERCISES 5.7

Exer. 1–12: Approximate the definite integral for the stated value of n by using (a) the trapezoidal rule and (b) Simpson's rule. (Approximate each $f(x_k)$ to four decimal places, and round off answers to two decimal places, whenever appropriate.)

1 $\int_1^3 (x^2 + 1) dx; \quad n = 4$

2 $\int_1^5 x^3 dx; \quad n = 4$

3 $\int_1^{1.6} (2x - 1) dx; \quad n = 6$

4 $\int_2^{3.2} (\frac{1}{2}x + 1) dx; \quad n = 6$

5 $\int_1^4 \frac{1}{x} dx; \quad n = 6$

6 $\int_0^3 \frac{1}{1+x} dx; \quad n = 8$

7 $\int_0^1 \frac{1}{\sqrt{1+x^2}} dx; \quad n = 4$

8 $\int_2^3 \sqrt{1+x^3} dx; \quad n = 4$

9 $\int_0^2 \frac{1}{4+x^2} dx; \quad n = 6$

10 $\int_0^{0.6} \frac{1}{\sqrt{4-x^2}} dx; \quad n = 6$

11 $\int_0^\pi \sqrt{\sin x} dx; \quad n = 6$

12 $\int_0^\pi \sin \sqrt{x} dx; \quad n = 4$

Exer. 13–16: Estimate the maximum error in approximating the definite integral for the stated value of n when using (a) the trapezoidal rule and (b) Simpson's rule.

13 $\int_{-2}^3 (\frac{1}{360}x^6 + \frac{1}{60}x^5) dx$; $n = 4$

14 $\int_0^3 (-\frac{1}{12}x^4 + \frac{2}{3}x^3) dx$; $n = 4$

15 $\int_1^5 \frac{1}{x^2} dx$; $n = 8$

16 $\int_1^4 \frac{1}{35}x^{7/2} dx$; $n = 6$

Exer. 17–20: Find the least integer n such that the error in approximating the definite integral is less than E when using (a) the trapezoidal rule and (b) Simpson's rule.

17 $\int_1^8 81x^{8/3} dx$; $E = 0.001$

18 $\int_1^2 \frac{1}{120x^2} dx$; $E = 0.001$

19 $\int_{1/2}^1 \frac{1}{x} dx$; $E = 0.0001$

20 $\int_0^3 \frac{1}{x+1} dx$; $E = 0.001$

Exer. 21–24: Suppose the table of values for x and y was obtained empirically. Assuming that $y = f(x)$ and f is continuous, approximate $\int_2^4 f(x) dx$ by means of (a) the trapezoidal rule and (b) Simpson's rule.

21

x	2.0	2.5	3.0	3.5	4.0
y	3	2	4	3	5

22

x	2.0	3.0	4.0
y	5	4	3

c 23

x	y
2.00	4.12
2.25	3.76
2.50	3.21
2.75	3.58
3.00	3.94
3.25	4.15
3.50	4.69
3.75	5.44
4.00	7.52

c 24

x	y
2.0	12.1
2.2	11.4
2.4	9.7
2.6	8.4
2.8	6.3
3.0	6.2
3.2	5.8
3.4	5.4
3.6	5.1
3.8	5.9
4.0	5.6

- c 25 Use the trapezoidal rule with $(b-a)/n = 0.1$ to show that

$$\int_1^{2.7} \frac{1}{x} dx < 1 < \int_1^{2.8} \frac{1}{x} dx.$$

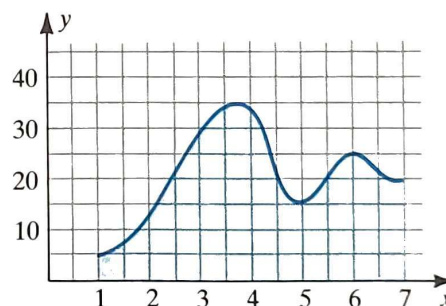
- c 26 It will follow from our work in Chapter 8 that

$$\int_0^1 \frac{1}{x^2 + 1} dx = \frac{\pi}{4}.$$

Use this fact and Simpson's rule with $n = 8$ to approximate π to four decimal places.

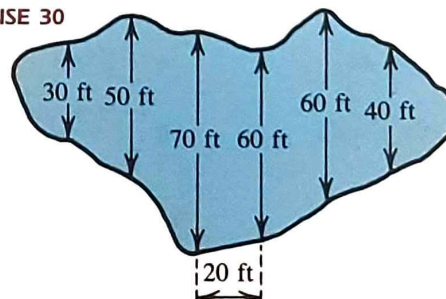
- 27 If $f(x)$ is a polynomial of degree less than 4, prove that Simpson's rule gives the exact value of $\int_a^b f(x) dx$.
- 28 Suppose that f is continuous and that both f and f'' are nonnegative throughout $[a, b]$. Prove that $\int_a^b f(x) dx$ is less than the approximation given by the trapezoidal rule.
- 29 The graph in the figure was recorded by an instrument used to measure a physical quantity. Estimate y -coordinates of points on the graph, and approximate the area of the shaded region by using (with $n = 6$) (a) the trapezoidal rule and (b) Simpson's rule.

EXERCISE 29



- 30 A man-made lake has the shape illustrated in the figure, with adjacent measurements 20 feet apart. Use the trapezoidal rule to estimate the surface area of the lake.

EXERCISE 30



- c 31 An important aspect of water management is the production of reliable data on *streamflow*, the number of cubic meters of water passing through a cross section of a stream or river per second. A first step in this computation is the determination of the average velocity $\bar{v}_x$ at a distance x meters from the river bank. If k is the

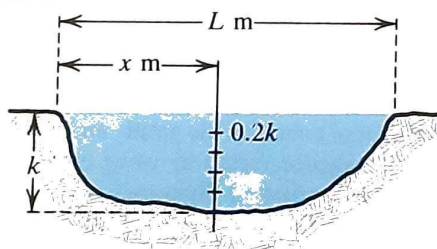
depth of the stream at a point x meters from the bank and $v(y)$ is the velocity (in m/sec) at a depth of y meters (see figure), then

$$\bar{v}_x = \frac{1}{k} \int_0^k v(y) dy$$

(see Definition (5.29)). With the *six-point method*, velocity readings are taken at the surface; at depths $0.2k$, $0.4k$, $0.6k$, and $0.8k$; and near the river bottom. The trapezoidal rule is then used to estimate $\bar{v}_x$. Given the data in the following table, estimate $\bar{v}_x$.

y (m)	0	$0.2k$	$0.4k$	$0.6k$	$0.8k$	k
$v(y)$ (m/sec)	0.28	0.23	0.19	0.17	0.13	0.02

EXERCISE 31



- [c] 32 Refer to Exercise 31. The streamflow F (in m^3/sec) can be approximated using the formula

$$F = \int_0^L \bar{v}_x h(x) dx,$$

where $h(x)$ is the depth of the stream at a distance x meters from the bank and L is the length of the cross section. Given the data in the following table, use Simpson's rule to estimate F .

x (m)	0	3	6	9	12
$h(x)$ (m)	0	0.51	0.73	1.61	2.11
$\bar{v}_x$ (m/sec)	0	0.09	0.18	0.21	0.36

x (m)	15	18	21	24
$h(x)$ (m)	2.02	1.53	0.64	0
$\bar{v}_x$ (m/sec)	0.32	0.19	0.11	0

- [c] Exer. 33–34: Use Simpson's rule with $n = 8$ to approximate the average value of f on the given interval.

33 $f(x) = \frac{1}{x^4 + 1}$; $[0, 4]$

34 $f(x) = \sqrt{\cos x}$; $[-1, 1]$

- [c] Exer. 35–36: If f is determined by the given differential equation and initial condition $f(0)$, approximate $f(1)$ using the trapezoidal rule with $n = 10$.

35 $f'(x) = \frac{\sqrt{x}}{x^2 + 1}$; $f(0) = 1$

36 $f'(x) = \sqrt{\tan x}$; $f(0) = 2$

5.8 REVIEW EXERCISES

Exer. 1–42: Evaluate.

- 1 $\int \frac{8x^2 - 4x + 5}{x^4} dx$
- 2 $\int (3x^5 + 2x^3 - x) dx$
- 3 $\int 100 dx$
- 4 $\int x^{3/5} (2x - \sqrt{x}) dx$
- 5 $\int (2x + 1)^7 dx$
- 6 $\int \sqrt[3]{5x + 1} dx$
- 7 $\int (1 - 2x^2)^3 x dx$
- 8 $\int \frac{(1 + \sqrt{x})^2}{\sqrt[3]{x}} dx$
- 9 $\int \frac{1}{\sqrt{x}(1 + \sqrt{x})^2} dx$
- 10 $\int (x^2 + 4)^2 dx$
- 11 $\int (3 - 2x - 5x^3) dx$
- 12 $\int (x + x^{-1})^2 dx$
- 13 $\int (4x + 1)(4x^2 + 2x - 7)^2 dx$
- 14 $\int \frac{\sqrt[4]{1 - (1/x)}}{x^2} dx$
- 15 $\int (2x^{-3} - 3x^2) dx$
- 16 $\int (x^{3/2} + x^{-3/2}) dx$
- 17 $\int_0^1 \sqrt[4]{8x^7} dx$
- 18 $\int_1^2 \frac{x^2 - x - 6}{x + 2} dx$
- 19 $\int_0^1 \frac{x^2}{(1 + x^3)^2} dx$
- 20 $\int_1^9 \sqrt{2x + 7} dx$
- 21 $\int_1^2 \frac{x + 1}{\sqrt{x^2 + 2x}} dx$
- 22 $\int_1^2 \frac{x^2 + 2}{x^2} dx$
- 23 $\int_0^2 x^2 \sqrt{x^3 + 1} dx$
- 24 $\int_1^1 3x^2 \sqrt{x^3 + x} dx$
- 25 $\int_0^1 (2x - 3)(5x + 1) dx$
- 26 $\int_{-1}^1 (x^2 + 1)^2 dx$
- 27 $\int_0^4 \sqrt{3x}(\sqrt{x} + \sqrt{3}) dx$
- 28 $\int_{-1}^1 (x + 1)(x + 2)(x + 3) dx$
- 29 $\int \sin(3 - 5x) dx$
- 30 $\int x^2 \cos(2x^3) dx$
- 31 $\int \cos 3x \sin^4 3x dx$
- 32 $\int \frac{\sin(1/x)}{x^2} dx$
- 33 $\int \frac{\cos 3x}{\sin^3 3x} dx$